

FACIAL BIOIMPEDANCE RESEARCH ROADMAP

*Task recognition to 3D EIT | 30 September
2026*

The research question is three different claims

A Few-electrode BioZ

Can permitted jaw/ear sites distinguish a preselected small task set across sessions and redonning?

Output: task label, not an image.

B 3D difference EIT

Can wider 3D electrode coverage localize coarse left/right facial-region changes and reject jaw/contact confounds?

Output: coarse change map + uncertainty.

C Individual muscle deficit

Can a validated map identify a weak muscle and quantify clinically meaningful impairment?

Output: calibrated biomarker. Highest risk.

More electrodes do not automatically yield 3D localization or a '% weak' muscle score.

What we measure: rest versus task

Rest state, v_0

Inject current. Sense boundary voltages. Skin, bone, blood, muscle, air, fluid and electrode contact are ALL present.

Task state, v_1

Repeat the SAME drive/sense protocol while the participant performs a task. Anatomy, geometry, contact and blood may all change.

$$\Delta v = v_1 - v_0 \neq \text{isolated muscle activation}$$

Interpret a task-dependent boundary pattern; one voltage sign does not identify a tissue or muscle.

Prior work map: what has and has not been shown

Study	Face?	Task recognition?	Image?	Key boundary
Liu 2025	Yes	Yes, cheek BioZ	No	n=4; anterior sites
Levit 2024	Yes	No; EMG + IPG	No	Vascular signal
Statland 2016	Yes	No; clinical EIM	No	Local OO probe
Schumann 2021	Yes	EMG task atlas	No EIT	Mixed activation
Ear EMG 2022	Ear	EMG task response	No EIT	Local ear signal
Kim 2019 / Piccin 2023	Lower face / neck	Airway tasks	2D / 3D airway	Not muscles
Our A → B → C	Target	To prove	To prove	Unobtrusive sites

P1 Liu et al. 2025 — Facial Gesture BioZ

1 What they did

Recognize cheek puff/suck gestures from facial bioimpedance, not image reconstruction.

2 Why difficult

Obtain task information without camera or dense facial EMG.

3 Their response

Place three contacts on the face, derive two BioZ channels, classify temporal patterns.

4 Method

20 Hz data; overlapping 20-sample windows (~1 s); Random Forest (100 trees); reported 80/20 split.

5 Setup

4 participants. One electrode below mouth; one on each cheek. Left/right/omni puff and suck; null also mentioned.

Full paper: <https://doi.org/10.1145/3745900.3746120>

P1 Liu et al. 2025 — Facial Gesture BioZ

6 Finding / conclusion

Mean accuracy 0.93 and macro-F1 0.92 in Table 1; task-dependent traces are visible.

Evidence boundary

n=4; anterior cheek electrodes; split grouping is not reported clearly; no across-session/subject proof; no muscle imaging.

7 Takeaway

Direct precedent: a few BioZ channels can encode facial task patterns. The numbers are not transferable to our harder placement.

8 Our project

Closest baseline for Paper A. Replicate task definitions where useful, but test jaw/ear sites, reapplication and held-out sessions.

Editor's reading

Puff/suck can change oral air and geometry; classification is not proof of isolated muscle conductivity.

Full paper: <https://doi.org/10.1145/3745900.3746120>

P2 Levit et al. 2024 — soft EMG + IPG

1 What they did

Simultaneously observe facial muscle-related biopotentials and vascular bioimpedance (IPG).

2 Why difficult

Facial artery and interwoven muscle responses are difficult to monitor concurrently without bulky contacts.

3 Their response

Screen-print flexible dry electrode arrays; use one for facial sEMG and another along superficial temporal artery.

4 Method

ICA on 16-contact biopotential array; four-contact IPG tracks impedance baseline and cardiac pulse.

5 Setup

4 healthy volunteers, ages 27–30; face and forearm comparisons; facial lip/teeth tasks and temporal artery region.

Full paper: <https://doi.org/10.1088/2057-1976/ad28cb>

P2 Levit et al. 2024 — soft EMG + IPG

6 Finding / conclusion

Muscle activation accompanied higher IPG baseline; artery narrowing is inferred by their model. Face pulse amplitude is less repeatable than forearm.

Evidence boundary

No direct vessel-radius measurement in every trial; n=4; local temporal-artery IPG, not EIT or task classifier.

7 Takeaway

Task–rest BioZ contains vascular and mechanical contributions. Baseline subtraction does not leave only muscle.

8 Our project

Add pulse/time controls; do not promise muscle-specific attribution from remote jaw/ear BioZ without validation.

Editor's reading

This paper supports our need to model blood vessels; it does not contradict Paper A's task-recognition aim.

Full paper: <https://doi.org/10.1088/2057-1976/ad28cb>

P3 Statland et al. 2016 — facial EIM in FSHD

1 What they did

Assess EIM reliability/validity in FSHD, including direct orbicularis oris measurement.

2 Why difficult

Smaller facial muscles, hair, and operator placement can degrade measurement repeatability.

3 Their response

Use a smaller handheld EIM array on bilateral orbicularis oris and multiple limb/trunk sites.

4 Method

Measure resistance/reactance/phase at 50 and 100 kHz; compare with severity/strength; test–retest ICC.

5 Setup

35 clinically affected FSHD participants; 18 retested within three weeks. Facial contact is local, not remote.

Full paper: <https://doi.org/10.1002/mus.25065>

P3 Statland et al. 2016 — facial EIM in FSHD

6 Finding / conclusion

Limb/trunk reliability generally high; facial right ICC <0.5 , left 0.84–0.91. Reactance related to disease metrics.

Evidence boundary

Disease=FSHD, not post-stroke dysarthria. No dynamic facial-task recognition, 3D EIT or per-muscle activation percentage.

7 Takeaway

Facial EIM is precedent, but reliability is a major challenge even with contact directly over target muscle.

8 Our project

Predefine redonning/session repeatability and contact-quality checks before clinical claims.

Editor's reading

Do not borrow its clinical correlations as proof that our ear/jaw classifier measures weakness.

Full paper: <https://doi.org/10.1002/mus.25065>

P4 Kusche et al. 2024 — dual-channel EIM

1 What they did

Build a wearable two-channel impedance spectrometer for real-time muscle-contraction detection.

2 Why difficult

Distinguish changing muscle impedance while accounting for frequency and electrode-interface error.

3 Their response

Tetrapolar local electrode pairs on two antagonistic forearm muscles; characterize instrument with phantom.

4 Method

Complex magnitude and phase across 20–230 kHz; 25 spectra/s/channel; R–C–R equivalent-circuit explanation.

5 Setup

Four electrodes per local muscle channel; phantom calibration and initial human forearm contractions.

Full paper: <https://doi.org/10.1109/JSEN.2024.3359284>

P4 Kusche et al. 2024 — dual-channel EIM

6 Finding / conclusion

Phantom statistical error $<1\%$ magnitude and $<0.4^\circ$ phase; initial forearm tasks show time/frequency-dependent responses.

Evidence boundary

Not a face model, not remote placement, not cohort-level recognition. Their optimum frequency is not ours.

7 Takeaway

Intra-/extracellular/membrane ideas describe frequency response; electrode contact and direction also matter.

8 Our project

Use frequency/phase and bench uncertainty only if CN0565 measurement chain supports them; validate on face separately.

Editor's reading

A circuit fitted to forearm is an effective model, not a literal map of individual facial cells.

Full paper: <https://doi.org/10.1109/JSEN.2024.3359284>

P5 Liu et al. 2024 — iMove

1 What they did

Recognize six upper-body fitness activities with wrist-to-wrist BioZ and wrist IMU.

2 Why difficult

Wearable motion tracking may miss body-state cues; BioZ alone is not necessarily strongest.

3 Their response

Fuse BioZ+IMU; contrastive training can improve a deployed IMU-only model.

4 Method

Leave-one-person-out evaluation; compare BioZ-only, IMU-only, fusion and BioZ-assisted training.

5 Setup

10 subjects across 5 days; electrodes at two wrists; left-wrist IMU; six activity classes.

Full paper: <https://doi.org/10.1109/PerCom59722.2024.10494489>

6 Finding / conclusion

Mean macro-F1: BioZ 75.36%, IMU 81.49%, fusion 89.57%, trained IMU-only 84.71%.

Evidence boundary

Upper-body movement, not face; conductive path spans the body and does not isolate a particular muscle.

7 Takeaway

A classifier can use distributed BioZ signals without reconstructing an image; modality ablations matter.

8 Our project

Paper A should test BioZ against video/motion baselines, not assume BioZ is always superior.

Editor's reading

Their cross-person split is a stronger evaluation idea than a random overlapping-window split.

1 What they did

Detect dining actions and food categories from wrist-to-wrist impedance.

2 Why difficult

Activity-associated impedance variations include changing external body–food conductive loops.

3 Their response

Treat those changes as useful features, not necessarily unwanted artifacts.

4 Method

Single two-electrode impedance channel; user-independent neural model; leave-one-person/meal-out analyses.

5 Setup

One electrode on each wrist; 10 volunteers, 40 meals, four intake activities and seven food types.

6 Finding / conclusion

Mean macro-F1 86.4% for four activities, 64.2% for seven food types.

Evidence boundary

Not facial muscle sensing and not EIT. The external food/utensil path is central to the method.

7 Takeaway

Task recognition may work through geometry/contact pathways even when muscle tissue properties are not isolated.

8 Our project

Decide whether oral-air/jaw/contact cues are legitimate for Paper A; exclude them only if claiming intrinsic muscle signal.

Editor's reading

A high classifier score cannot by itself explain the physiological mechanism.

P7 Schumann et al. 2021 — facial EMG atlas

1 What they did

Map relative activation patterns for 29 voluntary facial tasks, not measure BioZ.

2 Why difficult

Facial muscles overlap; isolated voluntary activation is uncommon, making single-muscle labels unreliable.

3 Their response

Record high-channel bilateral sEMG and display normalized activity on 3D facial illustrations.

4 Method

10 mimic muscle groups; 48-channel sEMG data; task-specific color normalization.

5 Setup

30 healthy men; 29 tasks including six German vowels, unilateral smiles, lip and cheek movements.

Full paper: <https://doi.org/10.1371/journal.pone.0254932>

P7 Schumann et al. 2021 — facial EMG atlas

6 Finding / conclusion

Most tasks co-activate several sampled muscles; purse lips and one-sided smile have dominant measured groups.

Evidence boundary

Buccinator not recorded; only men; surface EMG crosstalk; red is relative WITHIN each task, not cross-task strength.

7 Takeaway

Rename 'single-muscle anchor' to 'dominant-pattern task'. Use Table 1, not anatomy intuition alone.

8 Our project

Ground truth for task design and validation EMG; not proof BioZ differentiates those muscles.

Editor's reading

Five vowels are not inherited from this atlas: it measured six German vowels; choose language/IPA deliberately.

Full paper: <https://doi.org/10.1371/journal.pone.0254932>

P8 Rüschemschmidt et al. 2022 — ear EMG

1 What they did

Develop a method for measuring all nine intrinsic/extrinsic ear muscles during facial tasks.

2 Why difficult

Ear muscles are tiny and poorly mapped; facial tasks may activate them unexpectedly.

3 Their response

Cadaver anatomy guides electrode spots; compare needle and surface EMG, then test tasks/patients.

4 Method

Video-documented facial tasks; surface and selected needle EMG; unilateral patient comparison.

5 Setup

2 cadaver hemifaces; 12 healthy people; 7 patients with postparalytic facial synkinesis.

Full paper: <https://doi.org/10.3390/diagnostics12010121>

P8 Rüschemschmidt et al. 2022 — ear EMG

6 Finding / conclusion

Multiple ear muscles activate during most facial tasks; greatest measured activation with smiling.

Evidence boundary

This is EMG, not BioZ/EIT; condition is peripheral synkinesis, not post-stroke dysarthria.

7 Takeaway

Ear electrodes are not a biologically silent site. Ear-only recognition may rely on local ear co-activation.

8 Our project

Record auricular EMG in a validation subset or frame result as task recognition, not remote cheek-muscle imaging.

Editor's reading

Ear placement remains promising for unobtrusiveness, but its mechanism needs a direct control.

Full paper: <https://doi.org/10.3390/diagnostics12010121>

Adjacent EIT precedent: face, neck, partial coverage

Visentin 2017 — facial EIT

8 electrodes on one side of one face; exaggerated vowels/smile; deformation-inclusive proof of concept, not validated muscle map.

Kim et al. 2019 — lower-face airway

16 electrodes guided by MRI/landmarks; 10 healthy participants; compare EIT with MRI during swallowing-induced airway change.

Piccin et al. 2023 — 3D neck

32 electrodes in two bands; CT-derived 3D neck model; target is airway, not expression muscles.

What transfers to us

Use anatomy to target field coverage, test geometry and reconstruction. Their counts/placements do not prove our jaw/ear layout.

FULL LINKS

<https://tsukuba.repo.nii.ac.jp/record/43368/files/DA08090.pdf>

<https://doi.org/10.5664/jcsm.7714> | <https://doi.org/10.3389/frsle.2023.1238508>

Other facial BioZ / tissue-model precedents

Painometry 2020

Local corrugator-region impedance plus EEG/PPG/GSR for pain inference. Not remote facial-task recognition.

Schooling et al. 2020/21

Layered anisotropic tongue FEM + multifrequency Cole-type interpretation (41 ALS, 30 controls). Not facial expression.

A 2026 update

Büchner et al. use sEMG + 3D video (36 healthy adults) to visualize activation on a 3D face. That is NOT 3D EIT tomography.

FULL LINKS

<https://doi.org/10.1145/3386901.3389022> | <https://doi.org/10.1088/1361-6579/abcb9b>

<https://doi.org/10.1016/j.heliyon.2026.e45408>

Beginner's tissue model: where can Δv come from?

Component	Physical effect	Why it matters
Electrode + skin	Contact pressure/hydration	Can mimic task or drift
Fat / connective tissue	Depth and current path	Attenuates/redirects ROI sensitivity
Muscle cells	Extra-/intracellular + membrane	Frequency/anisotropy; contraction geometry
Blood vessels	Volume, pulse, constriction	Task-linked vascular BioZ (Levit)
Bone / teeth	Conductivity + geometry	Rest field and moving jaw
Mouth/nose air + saliva	Changing cavities/fluid paths	Strong cue in puff/suck/speech
Ear muscles	Local co-activation	Ear site is not physiologically silent

Each term can affect the voltage vector; anatomy at rest does not vanish after subtraction.

From equivalent circuit to 3D forward model

Material model (one effective tissue)

R_e : extracellular path

R_i : intracellular path

C_m : membrane effect

Frequency changes which pathways contribute.
Cole model summarizes a broad spectrum; fit parameters are not a named facial muscle.

Anatomical + electrode model

A 3D mesh assigns frequency-dependent conductivity/permittivity to skin, fat, muscle, bone, blood and cavities; a complete-electrode model handles contact.

Forward solve predicts boundary voltage for a chosen current pattern.

Material prior $\neq$ measured facial value. Use uncertainty ranges, anatomy registration, contact calibration and mesh-convergence checks.

<https://doi.org/10.1109/JSEN.2024.3359284> | <https://doi.org/10.1088/1361-6579/abcb9b> | <https://itis.swiss/virtual-population/tissue-properties/database/low-frequency-conductivity>

Placement depends on the question being asked

Layout family	Paper A: classifier	Paper B: 3D image	Specific control
Ear only	Wearable; test lower-face signal	Likely weak coverage; must test	Local ear EMG + head pose
Ear + jaw + submental	Candidate compromise	Maybe coarse lower-face ROI	Jaw/air/contact ablations
+ temple/forehead	More spatial diversity vs visibility	Potential multi-height coverage	Equal-count redistribution
Anterior cheek reference	Temporary Liu-like comparator	Improves target coverage; violates final constraint	Same device/protocol where possible

Closed ring is not a theorem. Partial coverage is possible but inverse localization must be measured.

Anatomy-informed example: <https://doi.org/10.5664/jcsm.7714>

The task list: correct the muscle labels

Proposed label	Atlas Table 1 says	Editorial correction
Purse lips (T14)	Orbicularis oris highest sampled	Dominant pattern, not isolated
Unilateral smiles (T22/23)	Zygomatic highest sampled	Separate left/right + video
Lower lip protrusion (T11)	Mentalis + depressors + OO	Mixed, not single target
Lips wide, jaw shut (T20)	Inferior OO + depressor labii + mentalis	Not depressor anguli highest
Upper-lip lift (T12)	Levators + OO	Mixed
Lip press / corners down	T7 / T8: multiple muscles	Mixed activation
Cheek puff / suck (T15/16)	OO + mentalis + depressor labii	Buccinator was NOT sampled

Source: Schumann et al., Table 1: <https://doi.org/10.1371/journal.pone.0254932>

Vowels, jaw and cheek: what are we actually testing?

Vowels

Atlas has SIX German vowels, not a universal five-vowel battery. Specify language, IPA and sustained vs speech production.

Jaw tasks

Jaw-open/closed-lips is atlas T18. Wide jaw opening and clench need separate rationale; clench is a jaw control, not a facial-muscle anchor.

Cheek puff/suck

Useful bridge to Liu, but air volume, oral pressure, lip seal and geometry can all change BioZ. Atlas did not measure buccinator.

Clinical relevance

Non-speech mimic tasks establish sensing feasibility; they do not by themselves prove dysarthria assessment or speech intelligibility.

Full links: <https://doi.org/10.1371/journal.pone.0254932> | <https://doi.org/10.1177/00220345770560071301>
<https://www.asha.org/Practice-Portal/Clinical-Topics/Dysarthria-in-Adults/>

Your 5 × 5 s protocol: calculated burden

If all 19 labels are used

6 anchor labels + 3 mixed + 5 vowels + 3 jaw + 2 cheek = 19.

5 repetitions each = 95 trials.

5 s active each = 475 s active time.

Timing depends on rest convention

With 5 s rest after every repetition: 475 s rest + 18×10 s between tasks = 1,130 s = 18 min 50 s.

Without final 5 s rest per task: 17 min 15 s.

Pilot timing, fatigue, signal recovery and task quality. These numbers are a proposal—not literature-validated BioZ parameters.

Recommended study design: prespecify before classifying

Primary classes

Rest, purse lips, left smile, right smile; optional lip press or cheek puff only with a clear use case.

Exploratory block

Vowels, jaw and cheek tasks test generalization/confounds; do not select classes after test accuracy.

Data and controls

Sync BioZ with video; EMG subset (including ear if needed); record contact, pose, jaw and pulse.

Evaluation

Split trials before windowing; report within-session, cross-session/redonning and unseen-person macro-F1 + uncertainty.

A valid first-paper claim

Constrained jaw/ear placement detects a preselected set of task patterns reliably under explicit nuisance controls. Do not claim individual-muscle imaging or post-stroke diagnosis.

Proof ledger: what must the next experiment decide?

Claim	Present evidence	Required proof	Status
Task changes facial BioZ	Liu, Levit	Replicate at permitted sites	Supported in general
Ear/jaw recognizes tasks	No exact-layout study	Matched tasks; held-out sessions	Hypothesis
A task isolates one muscle	Atlas shows co-activation	EMG if needed	False as stated
Δv is pure muscle	Blood/air/contact pathways	Nuisance controls + model	False as stated
3D localizes facial ROIs	Airway EIT precedent	3D FEM + ground-truth phantom	Unproven
Weakness percentage	No direct evidence	Defined scale + clinical validation	Future hypothesis

Parameter provenance: every number needs a reason

Parameter	Meaning	How to defend value
Electrode count / site	Coverage and burden	Equal-budget layout ablation + FEM
Drive/sense pattern	Independent boundary views	Rank/sensitivity + bench validation
Frequency / phase	Tissue dispersion and instrument	Verified hardware sweep; held-out ablation
Tissue σ , ϵ , anisotropy	Material properties	IT'IS/Gabriel range, uncertainty sweep
Contact + geometry	Boundary condition	Measure/reapply/scan + nuisance test
Trial timing / effort	Physiological repeatability	Pilot plateau, fatigue and recovery
Mesh h / regularization	Numerical/inverse assumptions	Convergence, phantom error, uncertainty

Prior: <https://itis.swiss/virtual-population/tissue-properties/database/low-frequency-conductivity>

Publication decision: staged, not promised

Paper A: feasible if signal survives

A few unobtrusive electrodes + selected task recognition + cross-session/redonning proof + clear mechanism limits.

Novelty is constrained placement and robust evaluation, not simply fewer classes.

Paper B: conditional and harder

Multi-plane/wider 3D EIT + coarse ROI reconstruction + identifiability + phantom truth.

Only then consider single-muscle and clinical weakness calibration as a separate claim.

First actionable gate: bench/phantom system verification and ethics/safety review before any human current injection.



Questions / decisions